.1 installed on or after 1 January 2000 that complies with:

.1 resolution MEPC.76(40), as amended …..…….....……■

.2 resolution MEPC.244(66) … .....……■

.2 installed before 1 January 2000 that complies with:

.1 resolution MEPC.59(33), as amended .....……..■

.2 resolution MEPC.76(40), as amended .....……..■

2.6 Equivalents (regulation 4)

The ship has been allowed to use the following fitting, material, appliance or apparatus to be

fitted in a ship or other procedures, alternative fuel oils, or compliance methods used as an

alternative to that required by this Annex:

System or equipment Equivalent used Approval reference

THIS IS TO CERTIFY that this Record is correct in all respects.

Issued at . . . .

. . . . . .

(place of issue of the Record)

-

Issued at . . . .

. . . . . .

(place of issue of the Record)

Date (dd/mm/yyyy) . . . .

. . . . . .

(date of issue) (signature of duly authorized

official issuing the Record)

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)

INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,

1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 56 -

Appendix II

Test cycles and weighting factors (regulation 13)

The following test cycles and weighting factors shall be applied for verification of compliance

of marine diesel engines with the applicable NOx limit in accordance with regulation 13 of this

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The following test cycles and weighting factors shall be applied for verification of compliance

of marine diesel engines with the applicable NOx limit in accordance with regulation 13 of this

Annex using the test procedure and calculation method as specified in the revised NO x

Technical Code 2008.

.1 For constant -speed marine engines for ship main propulsion, including

diesel-electric drive, test cycle E2 shall be applied.

.2 For controllable-pitch propeller sets test cycle E2 shall be applied.

.3 For propeller-law-operated m ain and propeller-law-operated auxiliary

engines the test cycle E3 shall be applied.

.4 For constant-speed auxiliary engines test cycle D2 shall be applied.

.5 For variable-speed, variable-load auxiliary engines, not included above, test cycle C1 shall be applied.

Test cycle for constant-speed main propulsion application (including diesel-electric drive and all controllable-pitch propeller installations)

Test cycle type

E2

-

cycle C1 shall be applied.

Test cycle for constant-speed main propulsion application (including diesel-electric drive and all controllable-pitch propeller installations)

Test cycle type

E2

Speed 100% 100% 100% 100%

Power 100% 75% 50% 25%

Weighting factor 0.2 0.5 0.15 0.15

Test cycle for propeller-law-operated main and propeller-law-operated auxiliary engine application

Test cycle type

E3

Speed 100% 91% 80% 63%

Power 100% 75% 50% 25%

Weighting factor 0.2 0.5 0.15 0.15

Test cycle for constant-speed auxiliary engine application

Test cycle type

D2

Speed 100% 100% 100% 100% 100%

Power 100% 75% 50% 25% 10%

Weighting factor 0.05 0.25 0.3 0.3 0.1

Test cycle for variable-speed and variable-load auxiliary engine application

Test cycle type

C1

Speed Rated Intermediate Idle

Torque 100% 75% 50% 10% 100% 75% 50% 0%

Weighting factor 0.15 0.15 0.15 0.1 0.1 0.1 0.1 0.15

-

Test cycle type

C1

Speed Rated Intermediate Idle

Torque 100% 75% 50% 10% 100% 75% 50% 0%

Weighting factor 0.15 0.15 0.15 0.1 0.1 0.1 0.1 0.15

In the case of an engine to be certified in accordance with paragraph 5.1.1 of regulation 13,

the specific emission at each individual mode point shall not exceed the applicable NO x

emission limit value by more than 50% except as follows:

.1 The 10% mode point in the D2 test cycle.

.2 The 10% mode point in the C1 test cycle.

.3 The idle mode point in the C1 test cycle.